

TECHNICAL NOTE

Context-Aware Action Evaluation on Modbus/TCP

An initial baseline in the OT Lab.

Overview

Liscere examines whether a protocol-valid and authorised industrial action is appropriate in its operational context. Protocol validity and access legitimacy are not sufficient to make that determination.

Modbus/TCP is used as the first baseline instantiation of the Liscere action-evaluation chain. This note reports an initial feasibility demonstration for one protocol and one process model. It shows that a decision can follow declared operational context, and not protocol validity alone.

The control pair

The pair holds the protocol operation, automation artefact, address, and value constant, and changes only the declared maintenance window. The same sensitive configuration write to the ALARM_HI_SP artefact returns two different decisions: an alert in production (maintenance_window false, OBS-R002), and an allow during a declared maintenance window (maintenance_window true, OBS-R004).

Scenarios and decisions

The maintenance window is not a blanket alert suppressor. An out-of-range setpoint write during maintenance still returns an alert, and a normal in-range write during maintenance remains an allow. All four constructed scenarios returned their expected decision and rule.

Scenario	Window	Decision	Evidence
Normal in-range write	true	ALLOW	OBS-R000
Out-of-range setpoint write	true	ALERT	OBS-R001
Sensitive write to ALARM_HI_SP	false	ALERT	OBS-R002
Sensitive write to ALARM_HI_SP	true	ALLOW	OBS-R004

What it shows, and what it does not

The demonstration shows that the action-evaluation chain runs end to end with one declared operational-context dimension, and that constructed scenarios can be checked against explicit expected decisions and evidence records. It does not measure detection performance, false-positive behaviour, robustness, or generality across context dimensions.

Scope and limitations

The context is declared, not inferred. The baseline covers one protocol and one controlled process model, with policy rules and artefact mappings defined manually. Multi-protocol validation remains future work, and the laboratory carries no production enforcement.

Availability

The scenarios run in OT Lab, the controlled laboratory. Source and scenarios are available at github.com/LiscereSecurity/OT-Lab.